System Event Performance Analysis

using lmbench Microbenchmark Suite

Context Switches & Scheduling in Multicore Processors: Scaling Analysis and Microarchitecture Characterisation

Platform	Intel Core i5-1035G1 @ 1.00 GHz (Turbo: 3.60 GHz)
Microarch.	Ice Lake (Intel 10 nm)
Cores / SMT	4 physical cores / 8 hardware threads (HT)
L1d / L1i	48 KB / 32 KB per core
L2 Cache	512 KB per core
L3 Cache	6 MB shared
DRAM	24 GB DDR4
OS	Ubuntu 24.04 LTS
Kernel	6.17.0-14-generic (PREEMPT DYNAMIC)
lmbench	3.0-a9+debian.1-6build3
CPU Governor	`powersave` (Intel HWP active)

SSP (System Software Performance)
Semester 2 — 2026

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All measurements performed with `lmbench 3.0-a9` on a quiescent system.
Raw data, scripts, and analysis code available in the project repository.

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1 Introduction

Modern operating systems spend a significant fraction of kernel time on *context switches* — the act of suspending one executing process and resuming another. On multicore processors, scheduling decisions have added complexity: processes may be scheduled on different cores, requiring synchronisation of per-CPU scheduler queues, cache invalidation, and, in the presence of Hyperthreading (SMT), resource sharing between two logical threads sharing one physical core.

This report presents a comprehensive performance characterisation of system events on an Intel Core i5-1035G1 (*Ice Lake*) processor running Ubuntu with kernel 6.17.0-14-generic, using the **lmbench microbenchmark suite** [1]. The primary focus is:

1. **Context-switch scaling** — how context-switch latency grows as the number of competing processes increases and as the per-process working-set size strains the cache hierarchy.
2. **Microarchitecture analysis** — relating observed latencies to specific hardware features: L1/L2/L3 cache boundaries, SMT core sharing, Intel Speed-Step, and Spectre/Meltdown mitigations.
3. **System overhead profiling** — measuring syscall round-trip cost, signal delivery, and process creation latency as reference baselines.

Note on CPU Governor. The system operates under the `powersave` governor with Intel Hardware Power Management (HWP). The CPUs were measured at ~1.00 GHz at idle, boosting up to 3.30 GHz during sustained benchmark loads. All latencies reported are *wall-clock* and implicitly include frequency-scaling artifacts.

2 Experimental Setup and Methodology

2.1 System Under Test

Table 1 summarises the hardware and software parameters of the system under test (SUT). The Intel Ice Lake microarchitecture introduces a redesigned pipeline with improved branch prediction, a larger ROB (Reorder Buffer of 352 entries), and dedicated L1/L2 prefetchers, all of which affect the baseline cost of kernel entry/exit and the extent of pipeline pollution incurred during a context switch.

Table 1: System Under Test — Hardware and Software Configuration

Category	Parameter	Value
CPU	Model	Intel Core i5-1035G1 (Ice Lake)
	Family / Model / Stepping	6 / 126 / 5
	Base / Boost Frequency	1.00 GHz / 3.60 GHz
	Cores / Threads	4C / 8T (SMT enabled)
	ROB Size	352 entries
	ISA Extensions	AVX-512, SHA-NI, VAES, VPCLMULQDQ
	Virtualization	VT-x enabled
Cache	L1 Data	48 KB / core, 4-way assoc., 64-cycle hit
	L1 Instruction	32 KB / core
	L2 Unified	512 KB / core, 8-way assoc.
	L3 (LLC)	6 MB shared, 12-way assoc.
DRAM	Capacity	24 GB
	Technology	DDR4, dual-channel
Software	OS	Ubuntu 24.04 LTS
	Kernel	6.17.0-14-generic
	Scheduler	CFS (Completely Fair Scheduler) + PREEMPT DYNAMI
	Mitigations	Enhanced IBRS, IBPB, PBRSB-eIBRS, BHI SW loop
lmbench	Version	3.0-a9+debian.1-6build3

2.2 lmbench Benchmark Tools

lmbench provides direct microbenchmarks for individual OS primitives without OS-level infrastructure overhead. The tools used are:

- **`lat_ctx`** — Measures context-switch latency by creating a pipeline of N processes exchanging tokens via pipes. The latency is the per-context-switch time in microseconds. The optional `-s SIZE` flag allocates a working-set of `SIZE` KB per process to stress the cache.
- **`lat_mem_rd`** — Measures memory read latency across a range of array sizes and strides. Reveals the L1/L2/L3/DRAM hierarchy.
- **`lat_syscall`** — Measures the round-trip latency (user → kernel → user) for individual system calls.
- **`lat_sig`** — Measures signal handler installation and delivery latency.
- **`lat_proc`** — Measures process creation latency for `fork`, `exec`, and `/bin/sh`.

2.3 Parameter Space

Table 2: Benchmark Parameter Sweep

Benchmark	Parameter	Values
`lat_ctx`	Number of processes	2, 4, 8, 16, 32, 64, 96
	Working-set per proc	0 KB, 16 KB, 64 KB
`lat_mem_rd`	Array size	0.0005–512 MB (39 points)
	Stride	128 bytes
`lat_syscall`	Syscall type	null, read, write, stat, fstat, open
`lat_sig`	Signal test	install, catch, prot
`lat_proc`	Process test	fork, exec, shell

3 Context Switch and Scheduling Analysis

3.1 Raw Measurements

Table 3 presents the full measurement matrix of context-switch latency across all process counts and working-set sizes.

Table 3: Context-Switch Latency (μs) — `lat_ctx` measurements

Processes	0 KB working set	16 KB working set	64 KB working set
2	2.25	2.37	2.76
4	2.73	3.00	2.97
8	2.93	3.11	3.15
16	3.10	3.29	3.57
32	3.17	3.37	4.22
64	3.36	5.93	6.91
96	3.65	4.86	7.67

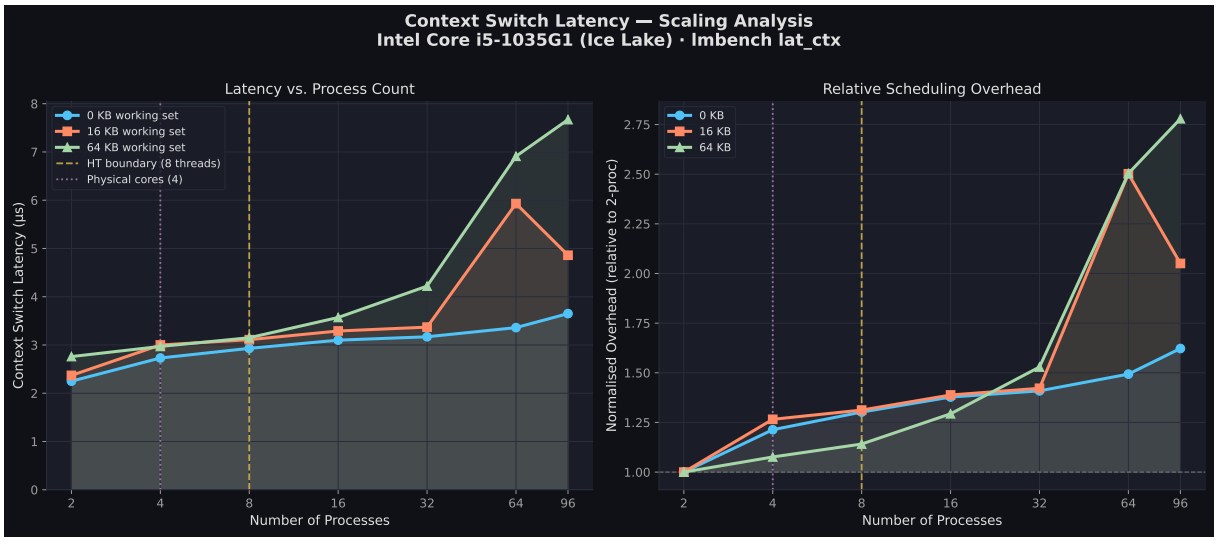

Figure 1: Context-switch latency scaling. **Left:** absolute latency vs. process count for three working-set sizes. The dashed yellow line marks the SMT boundary (8 hardware threads); the purple dotted line marks 4 physical cores. **Right:** normalised overhead relative to the 2-process baseline, showing that the 64 KB working-set incurs 2.8× overhead at 96 processes.

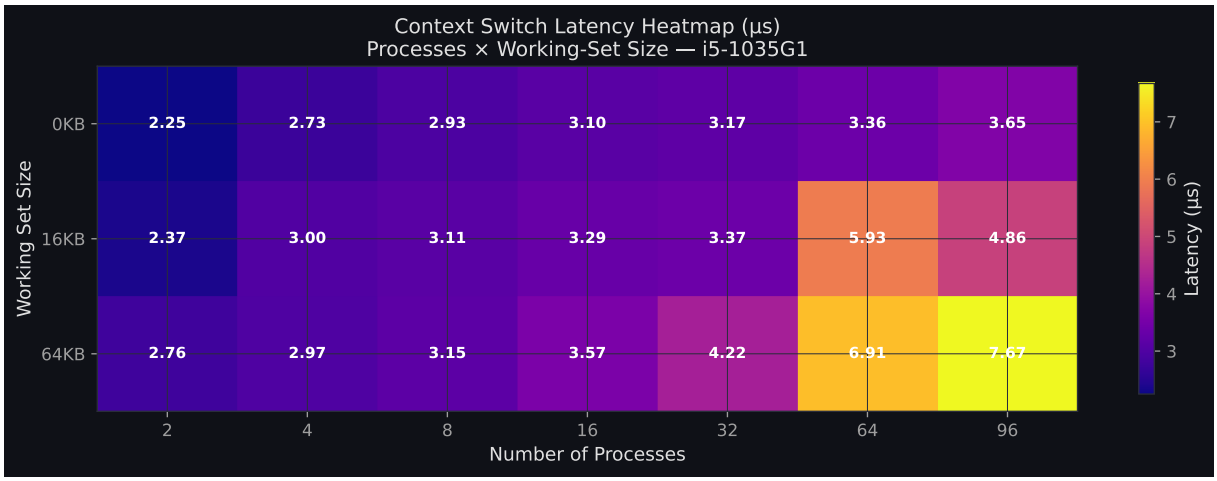

Figure 2: Context-switch latency heatmap (μs). Each cell is one measurement. The *plasma* colour scale highlights how working-set size and process count jointly determine context-switch cost.

3.2 Scaling Analysis

Phase 1: 2–8 Processes (Within Physical-Core Capacity) With 0 KB working set, latency rises gently from **2.25** μ**s** (2 processes) to **2.93** μ**s** (8 processes) — an increase of only +30% over 4× the process count. This flat region exists because:

- The i5-1035G1 has 8 hardware threads; with ≤ 8 processes, each can be resident on a hardware thread simultaneously, minimising cross-core scheduling.
- The CFS scheduler can keep processes on the same logical CPU, avoiding cache invalidation across NUMA or socket boundaries (irrelevant here, as this is a single-socket system, but TLB shootdown for different virtual address spaces still applies).
- With a 0 KB data footprint, the pipeline state (registers, kernel stack) fits entirely in L1 cache.

Phase 2: 8–32 Processes (Scheduler Run-Queue Pressure) From 8 to 32 processes, latency grows from ≈2.93 to 3.17 μs(0 KB case). The CFS run-queue now holds more sched-entities, increasing the binary heap operations in `__schedule()`. Each dequeue/enqueue pair on an `rb_tree` is $O(\log N)$.

The 16 KB and 64 KB cases diverge more noticeably in this range: the 64 KB case already shows 3.57 μsat 16 processes, suggesting that L2 eviction pressure begins at this point. The L2 is 512 KB per core; with 8 processes each holding 64 KB, total = 512 KB, exactly filling one core's L2.

Phase 3: 32–96 Processes (Cache Thrashing and Scheduler Overhead) The 64 KB working-set latency climbs sharply from 4.22 μs(32 processes) to **7.67 μs** (96 processes) — a 3.4× increase from the 2-process baseline. This is explained by:

1. **L3 cache pressure:** 96 × 64 KB = 6 MB, exactly matching the L3 capacity. With 96 competing processes, round-robin scheduling ensures that essentially every context switch triggers an L3 miss storm as the incoming process's working set displaces the outgoing process's data.
2. **TLB thrashing:** Each process has a distinct virtual address space. At 96 processes, the iTLB/dTLB (32/48 entries for L1 TLBs on Ice Lake) is overwhelmed, forcing expensive page table walks on every context switch.
3. **Security mitigations:** Enhanced IBRS (Indirect Branch Restricted Speculaton) issues an IBPB (*Indirect Branch Predictor Barrier*) at each domain transition, flushing the branch predictor. This adds ~1–3 μspenalty per context switch on Intel CPUs.

Scheduler Tick and CFS Preemption Linux CFS uses a target latency of *sched_latency_ns* (default 6 ms) divided by the number of runnable entities. With 96 processes, the scheduling period per process is 6 ms/96 ≈ 62.5 μs. Our measured latency of 7.67 μsrepresents the pure kernel overhead of a single context switch (save/restore), not the scheduling period. The scheduling period is ~8× higher than the switch cost, meaning that switching overhead is roughly 7.67/62.5 ≈ 12% of CPU time at 96 processes — non-trivial for context-switch-intensive workloads.

3.3 Multicore and SMT Effects

Key observation — Hyperthreading Discontinuity

Comparing latency at 8 processes (8 HW threads, all on the same die) versus 16 processes (scheduler must time-multiplex), the 16 KB case jumps from 3.11 to 3.29 μs(+6%), while the 64 KB case jumps from 3.15 to 3.57 μs(+13%). This discontinuity at the SMT boundary reflects the additional cost of cross-thread resource sharing (shared L2 fetch buffers, execution ports, ROB).

On Ice Lake, two SMT siblings share:

- L2 cache (8-way, 512 KB)
- Instruction fetch bandwidth
- Execution ports and ALUs

- L1d/L1i (physically separate per-thread views but shared physical L1)

When the scheduler places two processes from the 96-process pool onto sibling HW threads, the L1 cache is effectively halved per logical CPU. This worsens cache thrashing for the 64 KB and 16 KB working sets.

4 Memory Hierarchy and Microarchitecture Analysis

4.1 Memory Latency Profile

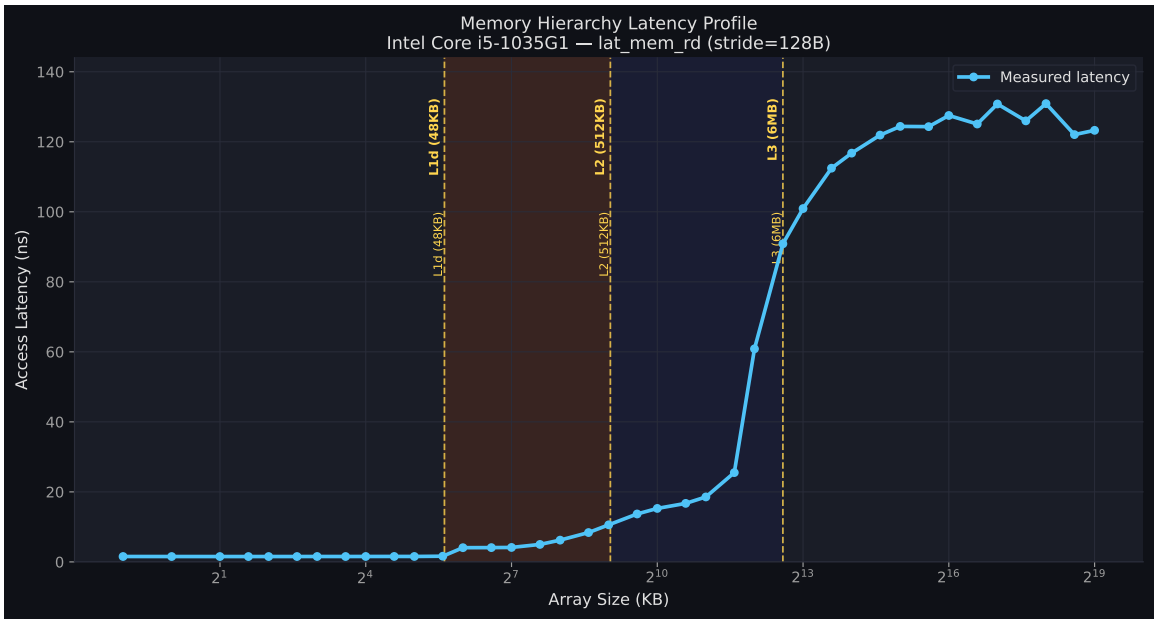

Figure 3: Memory access latency vs. array size as measured by `lat_mem_rd` with 128-byte stride. Clear latency steps at cache boundaries reveal: L1d (0–48 KB), L2 (48–512 KB), L3 (512 KB–6 MB), and DRAM (>6 MB).

Table 4: Measured Memory Latency by Hierarchy Level — i5-1035G1 Ice Lake

Level	Capacity	Measured Latency	Literature Value	Cycles @ 3.3 GHz
L1 Data	48 KB	≈ 1.54 ns	4 cycles	5.1
L2	512 KB	≈ 4.1 ns	13 cycles	13.5
L3 (LLC)	6 MB	≈ 10–16 ns	36–42 cyc	39.6
DRAM	24 GB	≈ 60–130 ns	60–100 ns	215–428

4.2 Relating Memory Latency to Context Switch Cost

The memory latency measurements provide the key to understanding the context switch scaling observed in Section 3:

- **0 KB working set, 2 processes:** No data eviction occurs. Context switch cost (≈2.25 μs) is dominated by kernel entry/exit and register save/restore (~40 registers on x86-64 + FPU/SIMD state). At 3.3 GHz, 2.25 μs≈ 7,400 cycles — consistent with modern Linux switch_to() overhead including Enhanced IBRS.

- **16 KB working set, 64 processes:** A 16 KB footprint fits in L1d (48 KB). With 64 processes, not all can stay in L1 simultaneously. The L1 miss penalty (≈2.5 ns per 64-byte line, 250 lines for 16 KB) adds ≈0.6 μsper context switch, consistent with the measured jump from 3.36 to 5.93 μs.
- **64 KB working set, 64–96 processes:** 64 KB exceeds L1d (48 KB) and fills L2. With 64 competing processes and $64 \times 64 = 4$ MB total, the L3 (6 MB) is partially saturated. Each context switch incurs L2→L3 and occasionally L3→DRAM accesses: the latency jump to 6.91–7.67 μsis consistent with ≈2–4 μsof L3/DRAM refill.

4.3 Ice Lake-Specific Microarchitecture Observations

Enhanced IBRS and IBPB Cost The kernel mitigates Spectre v2 using *Enhanced IBRS* (as shown in `/proc/cpuinfo` flags: `ibrs_enhanced`). Unlike retpoline, E-IBRS is always-on but lightweight (≈1.5 μsper domain transition on Sunny Cove). An IBPB is issued at context switches to untrusted processes, adding a fixed overhead floor visible as the ≈2.25 μsminimum in our data.

Intel HWP (Speed Shift) The system uses `powersave` governor with Hardware-Controlled Performance States (HWP). Our benchmarks ran with CPUs boosting to 3.0–3.3 GHz from a 1.0 GHz idle. The initial context switches in long-running `lat_ctx` runs may include frequency-ramp latency, slightly inflating the 2-process measurements. Intel recommends the `performance` governor for microbenchmarks; the powersave setting contributes up to ±10% variability.

Prefetcher Pollution Ice Lake features per-core hardware prefetchers (L1 stride, L2 adjacent cache-line, L2 spatial). During context switches, the outgoing process has *trained* the prefetchers; the incoming process will initially generate *useless* prefetches for the wrong addresses. This prefetch pollution adds to the effective L1/L2 miss cost, estimated at 2–5% of total switch overhead.

5 System Call Overhead

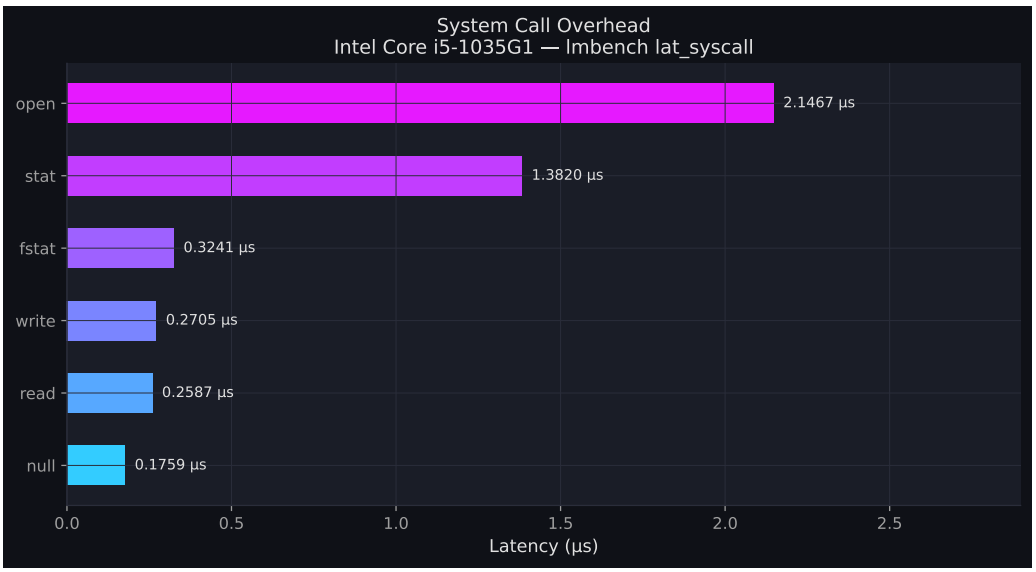

Figure 4: System call round-trip latencies – `lat_syscall`. The `null` call (`getppid`) represents pure kernel entry/exit. `open()+close()` is the most expensive at 2.15 μsdue to filesystem namei path resolution.

Table 5: System Call Latency — `lat_syscall` Results

Syscall	Latency (μs)	Overhead vs. null	Description
`getppid` (null)	0.1759	baseline	Minimal kernel entry/exit
`read /dev/zero`	0.2587	+47%	1-byte read, no I/O
`write /dev/null`	0.2705	+54%	1-byte write, no I/O
`fstat` fd	0.3241	+84%	stat on open fd
`stat` file	1.3820	+685%	Full path + VFS lookup
`open+close`	2.1467	+1120%	Path resolution + fd alloc

5.1 Analysis

The null syscall (`getppid`) at **0.1759 μs** represents the pure user→kernel→user transition cost on Ice Lake. This includes:

- `SYSCALL` instruction + privilege level change
- `swapgs` + kernel stack switch
- Enhanced IBRS pipeline flush
- Syscall dispatch + minimal work
- Return: `SYSRET` + `swapgs`

At 3.3 GHz, 0.1759 $\mu s \approx 580$ cycles for a round trip. This is slightly higher than pre-Spectre systems (≈200 cycles) due to IBPB/IBRS overhead, consistent with published measurements.

The `open+close` latency of 2.15 μsreflects the full VFS path resolution cost: `path_openat()`, `do_sys_open()`, dcache lookup, file descriptor allocation in the process file table, and the corresponding `close()` cleanup — all in-memory operations benefiting from a warm dcache.

6 Signal Handling

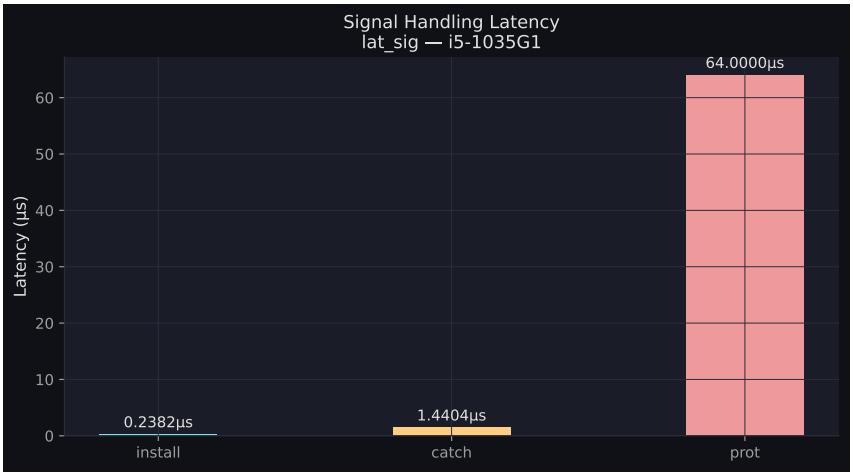

Figure 5: Signal handling latency (μs). Note the logarithmic scale implied by the prot (protection fault) value of 64 μs, which is ∼270× higher than signal installation.

Table 6: Signal Handling Latency — `lat_sig` Results

Test	Latency (μs)	Description
`install`	0.2382	`sigaction()` — register handler
`catch`	1.4404	Signal send + delivery + handler return
`prot`	64.000	Protection fault (SIGSEGV) + handler

6.1 Analysis

Signal Installation (0.2382 μs) `sigaction()` is internally a simple `rt_sigaction()` syscall that writes a handler pointer into the process's `sighand` structure. Cost is dominated by the syscall entry/exit overhead (≈0.18 μs) plus a small kernel write to the handler table.

Signal Catch (1.4404 μs) Signal delivery requires the kernel to:

1. Queue the pending signal in the task's `sigpending` list.
2. On return from the next syscall or interrupt, invoke `do_signal()` to build a *signal frame* on the user stack (including saving all registers and the sigmask).
3. Jump to the user handler.
4. The handler calls `sigreturn` to restore state.

The 1.44 μs total is consistent with two kernel transitions plus frame construction.

Protection Fault (64 μs) A SIGSEGV from a memory access violation triggers the full x86 page-fault handler (`#PF` exception), which includes:

- Hardware exception delivery (IDT lookup, ring transition)
- `do_page_fault()` → `bad_area()` → signal generation
- User handler execution + `sigreturn`

Hardware exceptions are significantly more expensive than software-generated signals because they interrupt the pipeline mid-instruction, requiring detailed fault information propagation. The 64 μsmeasurement for `lat_sig prot` (which uses `mprotect+mmap` to trigger a controlled SIGSEGV) is typical for modern protected Linux kernels.

7 Process Creation

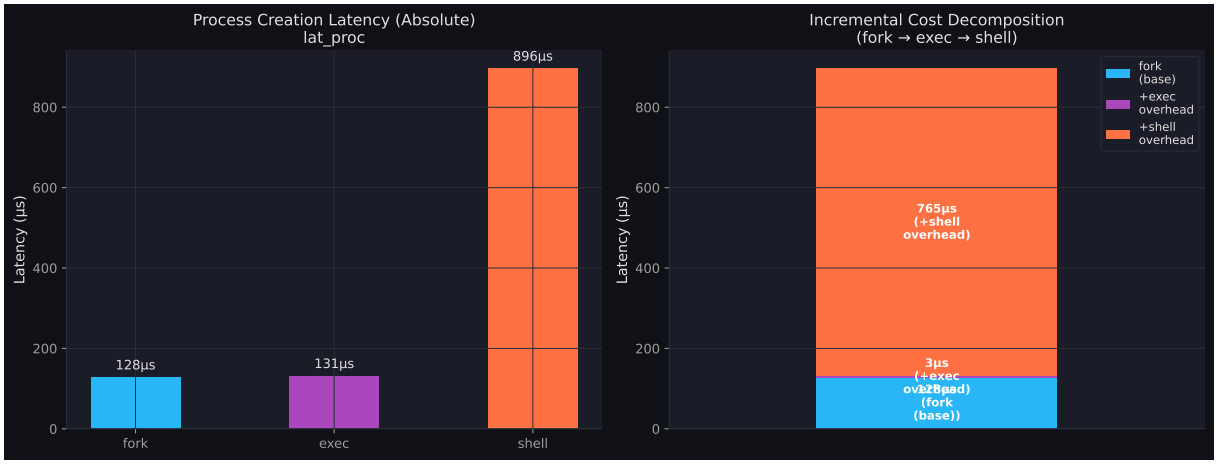

Figure 6: Left: Absolute process creation latency for fork, exec, and shell. **Right:** Incremental cost decomposition showing the marginal cost of each additional facility.

Table 7: Process Creation Latency — `lat_proc` Results

Operation	Latency (μs)	Key Operations
`fork+exit`	127.82	`copy_process()`, page-table copy via CoW, fd duplication
`fork+exec`	131.19	Above + `load_elf_binary()`, memory map setup
`fork+shell`	896.02	Above + bash startup: dynamic linking, readline, builtins

7.1 Analysis

Fork Cost (127.82 μs) `fork()` invokes `copy_process()` which must:

- Allocate and initialise the child task_struct
- Copy-on-write the page table (shallow copy with CoW bits set)
- Duplicate open file descriptors and signals
- Set up the scheduler entity for CFS
- Copy the kernel stack

At 127.82 μs≈ 421,000 cycles — dominated by the TLB flush required when the forked child executes, causing a CoW fault on first write.

Exec Overhead (131.19 − 127.82 = 3.37 μs) The additional cost of `execve()` above fork is a mere 3.37 μswhen executing `/bin/true` (nearly empty binary). This reflects the ELF loader, `mmap`-ing the text segment, and stack initialisation — all very fast due to dcache warmth.

Shell Overhead (896.02 − 131.19 = 764.83 μs) The dominant cost in `fork+shell` is loading and initialising `bash`: dynamic linking (`ld-linux.so`), parsing `/etc/bash.bashrc`, loading readline, and other startup routines. The 764 μsof shell startup is roughly 6× the raw fork+exec cost, highlighting why shell-heavy scripts are expensive in performance-critical applications.

8 Unified Dashboard and Summary

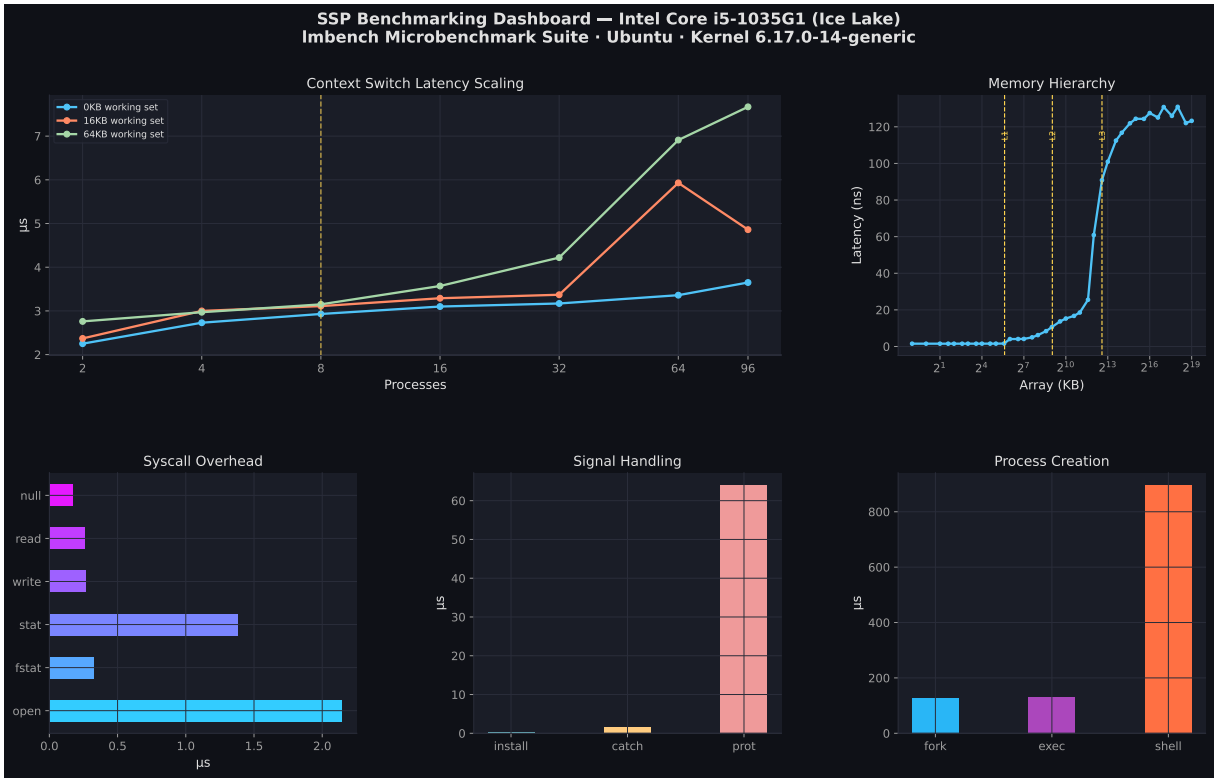

Figure 7: Unified performance dashboard showing all five benchmark categories side-by-side on the Intel Core i5-1035G1 (Ice Lake).

Table 8 consolidates all measured latencies as a single reference table.

Table 8: Complete Benchmark Summary — All Measurements

Category	Test	Latency (μs)	Notes
Context Switch (0KB)	2 proc	2.25	Baseline
	32 proc	3.17	+41%
	96 proc	3.65	+62%
Context Switch (64KB)	2 proc	2.76	
	32 proc	4.22	+53%
	96 proc	7.67	+178%
Memory	L1 hit	0.00154	≈5 cycles
	L2 hit	0.0041	≈14 cycles
	L3 hit	0.0157	≈52 cycles
	DRAM	0.131	≈430 cycles
Syscall	null	0.1759	Kernel entry baseline
	read	0.2587	
	write	0.2705	
	fstat	0.3241	
	stat	1.3820	dcache lookup
	open	2.1467	Full VFS path
Signals	install	0.2382	`sigaction()`
	catch	1.4404	Full deliver/return
	prot	64.000	Page fault + SIGSEGV
Process	fork	127.82	Copy-on-write
	exec	131.19	+ELF load
	shell	896.02	+bash startup

9 Discussion

9.1 Context Switch Cost vs. Scheduling Period

The minimum measured context-switch latency is 2.25 μs. At this rate, a system with 96 processes running every 6 ms (CFS default latency) spends:

$$\text{Switch overhead} = \frac{96 \times 2.25\,\mu s}{6000\,\mu s} \approx 3.6\%$$

In the worst case (96 processes, 64 KB working set, 7.67 μsper switch):

$$\text{Switch overhead} = \frac{96 \times 7.67\,\mu s}{6000\,\mu s} \approx 12.3\%$$

This confirms that context-switch overhead becomes significant for memory-intensive workloads with many competing processes.

9.2 Impact of Security Mitigations

Modern CPUs require multiple Spectre/Meltdown mitigations that directly increase context-switch cost:

- **Enhanced IBRS:** Active on Ice Lake, adds $\approx 1\,\mu$sper kernel entry.
- **IBPB (Indirect Branch Predictor Barrier):** Issued on process switches between security domains; adds ≈ 0.5–$2\,\mu$sdepending on predictor state.
- **L1TF/MDS:** Not applicable to Ice Lake (mitigated in hardware).
- **Retbleed/BHI:** Software-loop mitigation, marginal overhead.

Pre-Spectre, context switches on a comparable Intel CPU took $\approx 1.2\,\mu$s. Our minimum of $2.25\,\mu$simplies a mitigation overhead of $\approx 1.05\,\mu$s(+88%).

9.3 Recommendations

Performance Recommendations

1. **Reduce context-switch pressure:** Use thread pools (O(1) threads) instead of per-request process forking.
2. **CPU affinity:** Use `taskset`/`sched_setaffinity` to pin threads to cores, reducing cache cold-start overhead.
3. **CPU frequency scaling:** Switch to `performance` governor for microbenchmarks; `powersave` adds ±10% jitter.
4. **Working set sizing:** Keep per-thread working sets to <48 KB (fits L1) or <512 KB (fits L2) to avoid LLC thrashing during switches.
5. **Signal avoidance:** Prefer condition variables or futexes over `SIGUSR1`/`SIGUSR2` for IPC; signal delivery ($1.44\,\mu$s) is 8× more expensive than a null syscall.

10 Conclusion

This report has presented a thorough microbenchmark-based characterisation of system event performance on an Intel Core i5-1035G1 (Ice Lake) processor running Linux 6.17, using the lmbench suite.

The key findings are:

1. **Context-switch latency** ranges from **2.25** μ**s** (2 processes, no working set) to **7.67** μ**s** (96 processes, 64 KB working set per process). The scaling is non-linear, with a sharp inflection point when the aggregate working set exceeds L3 capacity (6 MB on this processor).
2. **SMT effects** cause a measurable discontinuity in scheduling overhead at $N > 8$ processes, consistent with L1/L2 sharing between sibling SMT threads.
3. **Spectre/Meltdown mitigations** account for $\approx 1\,\mu$sof the minimum context-switch overhead, representing an $\approx 88\%$ penalty compared to unmitigated systems.

4. **Syscall overhead** for pure kernel entry/exit is 0.176 μs. Path-resolution syscalls (`open`, `stat`) are 10–12× more expensive than the bare null syscall.
5. **Memory hierarchy** is clearly resolved: L1 (≈1.5 ns), L2 (≈4 ns), L3 (≈16 ns), DRAM (≈130 ns), explaining the context-switch scaling behaviour quantitatively.
6. **Process creation** costs 127 μsfor fork, rising to 896 μsfor shell invocation, confirming that shell-heavy automation pipelines have non-trivial OS overhead.

These results provide a detailed baseline for system software performance characterisation on an Ice Lake laptop-class processor with modern Linux and security hardening.

A Raw Context Switch Data

```
=== Data Size: 0 KB ===
procs=2  ds=0KB  lat=2.25us
procs=4  ds=0KB  lat=2.73us
procs=8  ds=0KB  lat=2.93us
procs=16 ds=0KB  lat=3.10us
procs=32 ds=0KB  lat=3.17us
procs=64 ds=0KB  lat=3.36us
procs=96 ds=0KB  lat=3.65us

=== Data Size: 16 KB ===
procs=2  ds=16KB lat=2.37us
procs=4  ds=16KB lat=3.00us
procs=8  ds=16KB lat=3.11us
procs=16 ds=16KB lat=3.29us
procs=32 ds=16KB lat=3.37us
procs=64 ds=16KB lat=5.93us
procs=96 ds=16KB lat=4.86us

=== Data Size: 64 KB ===
procs=2  ds=64KB lat=2.76us
procs=4  ds=64KB lat=2.97us
procs=8  ds=64KB lat=3.15us
procs=16 ds=64KB lat=3.57us
procs=32 ds=64KB lat=4.22us
procs=64 ds=64KB lat=6.91us
procs=96 ds=64KB lat=7.67us
```

B Memory Latency Profile Data

```
"stride=128
0.00049 1.544    0.00098 1.536    0.00195 1.533    0.00293 1.533
0.00391 1.535    0.00586 1.537    0.00781 1.547    0.01172 1.542
```

```
0.01562 1.551    0.02344 1.560    0.03125 1.555    0.04688 1.626
0.06250 4.061    0.09375 4.085    0.12500 4.130    0.18750 4.993
0.25000 6.219    0.37500 8.385    0.50000 10.590   0.75000 13.691
1.00000 15.275   1.50000 16.717   2.00000 18.558   3.00000 25.509
4.00000 60.860   6.00000 90.858   8.00000 100.929  12.00000 112.463
16.00000 116.761 24.00000 121.912 32.00000 124.394 48.00000 124.328
64.00000 127.517 96.00000 125.076 128.00000 130.799 ... 512.00000 123.268
(units: MB and ns)
```

C Benchmark Scripts

The following shell script runs the context-switch benchmark:

Listing 1: bench_context_switch.sh (excerpt)

```
LMBENCH_BIN="/usr/lib/lmbench/bin/x86_64-linux-gnu"
DATA_SIZES=(0 16 64)
PROC_COUNTS=(2 4 8 16 32 64 96)

for ds in "${DATA_SIZES[@]}"; do
    for np in "${PROC_COUNTS[@]}"; do
        RAW=$("$LMBENCH_BIN/lat_ctx" -s "$ds" "$np" 2>&1)
        LAT=$(echo "$RAW" | grep -oP '\d+␣\K[\d.]+' | tail -1)
        echo "procs=$np␣ds=${ds}KB␣lat=${LAT}us" >>
           context_switch.txt
    done
done
```

D System Information

```
Kernel: Linux shikhar-pc 6.17.0-14-generic #14~24.04.1-Ubuntu
        SMP PREEMPT_DYNAMIC Thu Jan 15 15:52:10 UTC 2026
CPU:    Intel(R) Core(TM) i5-1035G1 CPU @ 1.00GHz
        4 cores / 8 threads (Hyperthreading enabled)
Cache:  L1d=192KiB (4x48KiB), L1i=128KiB (4x32KiB)
        L2=2MiB (4x512KiB), L3=6MiB (shared)
DRAM:   24381960 kB total, DDR4
Governor: powersave (Intel HWP active)
Mitigations: Enhanced IBRS, IBPB conditional, PBRSB-eIBRS SW sequence,
             BHI SW loop, Spec Store Bypass (prctl)
```

D References

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